

Packet Tracer - Lab 26 - Network Representation

Objectives

The network model in this activity incorporates many of the technologies that you will master in your CCNA studies. It represents a simplified version of how a small to medium-sized business network might look. Feel free to explore the network on your own. When you are ready, proceed through the following steps and answer the questions.

Note: It is not important that you understand everything you see and do in this activity. Feel free to explore the network on your own. If you wish to proceed more systematically, follow the steps below. Answer the questions to the best of your ability.

Instructions

Step 1: Identify common components of a network as represented in Packet Tracer.

The icon toolbar at the bottom left hand corner has various categories of networking components. You should see categories that correspond to intermediary devices, end devices, and media. The **Connections** category (with the lightning bolt icon) represents the networking media supported by Packet Tracer. There is also an **End Devices** category and two categories specific to Packet Tracer: **Custom Made Devices** and **Multiuser Connection**.

List the intermediary device categories. **Router, Switches, Hubs, Wireless Devices, Security, WAN Emulation**

Without entering into the internet cloud or intranet cloud, how many icons in the topology represent endpoint devices (only one connection leading to them)? **15**

Without counting the two clouds, how many icons in the topology represent intermediary devices (multiple connections leading to them)? **13**

How many end devices are **not** desktop computers? **8**

How many different types of media connections are used in this network topology? **4 (Serial, Coaxial, Copper Straight, WiFi)**

Step 2: Explain the purpose of the devices.

a. In Packet Tracer, only the Server-PT device can act as a server. Desktop or Laptop PCs cannot act as a server. Based on your studies so far, explain the client-server model. **The client is a machine that makes requests for information or resources and the Server is the device machine that provides the information or resources to the client requester.**

b. List at least two functions of intermediary devices. **Forward information or resources from servers or other intermediary devices to end client devices and vice versa (forward requests for information or resources from end client devices to servers or other intermediary devices).**

c. List at least two criteria for choosing a network media type. **Type of data being transmitted and the distance between nodes needing the data being transmitted.**

Step 3: Compare and contrast LANs and WANs.

- a. Explain the difference between a LAN and a WAN. Give examples of each. **A LAN is a local area network of devices like modem, layer 2 switches, and devices for communicating within the LAN. Whereas the WAN is a wide area network of LANs and uses routers and layer 3 switches to connect LAN or WAN to one another.**
- b. In the Packet Tracer network, how many WANs do you see? **3**
- c. How many LANs do you see? **4**
- d. The internet in this Packet Tracer network is overly simplified and does not represent the structure and form of the real internet. Briefly describe the internet. **There is an ISP Tier 3a router that links to its local ISP Tier 3b switch that connects to its 2 servers: famous.dns.pka and netacad.pka. A cable provider and 2 business WANs are connected to the Tier 3a router.**
- e. What are some of the common ways a home user connects to the internet? **Cable Provider, from ISP provides access so clients can use browsers, FTP**
- f. What are some common methods that businesses use to connect to the internet in your area? **Direct connect from business router to Tier 3a ISP provides access so clients can use browsers, FTP**

Challenge Question

Now that you have had an opportunity to explore the network represented in this Packet Tracer activity, you may have picked up a few skills that you would like to try out. Or maybe you would like the opportunity to explore this network in more detail. Realizing that most of what you see and experience in Packet Tracer is currently beyond your skill level, here are some challenges you might want to attempt. Do not worry if you cannot do them all. You will be a Packet Tracer master user and network designer soon enough.

1. Add an end device to the topology and connect it to one of the LANs with a media connection.
 - a. What else does this device need to send data to other end users? **Needed to add an IP address, subnet mask, and default gateway**
 - b. Can you provide the information? **Yes, can be identified by looking at other device configurations in the LAN**
 - c. Is there a way to verify that you correctly connected the device? **Yes, ping other devices in the subnet, then ping the ISP Tier 3a router.**
2. Add a new intermediary device to one of the networks and connect it to one of the LANs or WANs with a media connection. What else does this device need to serve as an intermediary to other devices in the network? **Just a connection to the other intermediary devices**
3. Open a new instance of Packet Tracer. Create a new network with at least two LANs connected by a WAN. Connect all the devices. Investigate the original Packet Tracer activity to see what else you might need to do to make your new network functional.
 - a. Record your thoughts and save your Packet Tracer file. You may want to revisit your network later after you have mastered a few more skills.

